

# **Social Media Algorithm and Political Polarization**

**Ankush Sharma, Bhupesh Relan, Chirag Singla, Anuj Anand**

([ankush1283.be22@chitkara.edu.in](mailto:ankush1283.be22@chitkara.edu.in),[bhupesh1447.be22@chitkara.edu.in](mailto:bhupesh1447.be22@chitkara.edu.in),[chirag1466.be22@chitkara.edu.in](mailto:chirag1466.be22@chitkara.edu.in),[anuj1315.be22@chitkara.edu.in](mailto:anuj1315.be22@chitkara.edu.in))

**Department of Computer Science & Engineering**

**Chitkara University, Punjab, India**

## **Abstract:**

The algorithms that rule your social media feeds determine what content you'll be able to see on platforms and apps. The ultimate goal of these algorithms is to keep users engaged by surface more relevant content, i.e. content you are likely to like, comment on, or share. But when it comes to political discourse online, this system can have its downsides. Since emotional, shocking, or contentious content drives engagement, that kind of content is what typically gets promoted to users. This can result in a "filter bubble," a feed full of information that agrees with and reinforces the user's current beliefs.

Algorithms on social media can create echo chambers in politics by limiting the amount of opposing views users see online. In the long run, this can make online politics even more polarized and closed off. In this study we explore the relationship between algorithms and (polarizing) politics. The first part of the study maps current research in this area and situates it within disciplines including (applied) political science and media studies. This research however goes beyond a traditional literature review and also includes the training and testing of a machine learning model that processes and analyzes online social media data to uncover insights in the spread of political information within online social networks. Specifically we seek to understand what information is spread and to what extent users adopt a range of perspectives and comments within political threads.

New research found that algorithms that seek to maximize engagement, tend to surface more extreme, outrageous, or emotive content because it receives the most interactions. This can have the effect of reinforcing users existing views, reducing the potential for exposure to opposing perspectives, and accelerating the spread of misinformation. The analysis also found that users who regularly interact with political content on Twitter are increasingly likely to see the same types of content in their timeline.

Research has found that algorithms may be perpetuating personalisation of content online – and exacerbating political polarization. To address these challenges, our research calls for greater algorithmic transparency, the development and enforcement of new regulations for online intermediaries and enhanced digital literacy to better equip citizens to negotiate and critically consume online information.

**Keywords:** Algorithm Bias, Political Polarization, filter bubbles, Engagement Driven Content, Echo chamber

## **Overview:**

The algorithms that power social media sort through the deluge of content on your feed by prioritizing the things that your friends and followers have engaged with in the past. Because most content gets sorted by past engagement, social media algorithms tend to sort political content by past engagement as well, meaning that people see posts that are similar to the ones they already agree with. Because certain types of political content, like deeply emotional or extreme statements, elicit more emotional responses than others, these kinds of posts get prioritized. This study uses machine learning techniques and analysis of real-world social media datasets to better understand how engagement algorithms on social media can lead to the spread of misinformation and increased polarization. But there is hope. By increasing transparency, regulating features that contribute to the spread of misinformation, and educating users on how to effectively use social media, social media algorithms can be designed to reduce, rather than increase, political divisions.

## **Introduction:**

Social media platforms are fast becoming one of the primary sources where individuals access news and are informed about political issues. Not only can individuals read updates as they appear online, they are also able to share their own thoughts and opinions, whilst engaging in real time conversation with others around the world. On any given day there is an incredible amount of content posted online, therefore it is unrealistic to assume individuals will be able to see every update that is relevant to them. Instead, algorithms are used to curate content and display updates considered relevant by the algorithm.

The primary function of most of the algorithms is to keep individuals engaged. By analyzing user behavior, the algorithms seek to determine what users are searching for, how much time a user spends watching a particular video, and what types of videos and posts users like to share, comment on, and over all engage with. The algorithm attempts to provide users with information that the user would find relevant and of interest to them. Although this is an effective and time saving tool for the average individual, it also allows individuals to access a vast amount of political information. There is a growing concern that content which is more emotive, dramatic and based on personal opinions is more likely to garner attention from algorithms. As a result, users are only likely to encounter viewpoints that are similar to their own opinion, which can create so-called echo chambers where people become more solidified in their views and, potentially, more extreme. And this is leading to further division within the political sphere. The problem is also compounded by the spread of false information through algorithms that focus more on engagement and viral sharing than on accuracy. This could have profound implications for our ability to consume accurate information, and consequently, our thought processes, particularly in times of great social change and during elections. In today's information age there exists a significant amount of political content disseminated through social media, while people have become increasingly polarized when it comes to politics. In this research, we will look at current studies and cases about algorithms found in today's social media, and use a machine learning approach to look into the information people see and engage in on social media. Our ultimate goal is to analyze how social media algorithms affect political polarization, and, most importantly, how to reverse its negative effects.

### **i. Background on Social Media Algorithm:**

Social media platforms are now mainly automated in order to cope with the volume of content users are uploading on a daily basis. Facebook, YouTube and Twitter are the main platforms that use algorithms, and this is because they are unable to physically display every single post to every single user. The algorithm sorts the content and ranks it based on things such as watch time, likes, shares, comments and more, and the main intention behind this is to increase overall engagement. Users will see posts that they may be interested in, and the algorithm continuously adapts in order to keep users on the site for longer periods of time.

How do the social media giants make those decisions about what you see in your feed? Machine learning models trained on massive amounts of user data power the decision-making for Facebook, Instagram, Twitter, LinkedIn, and others. Those models look for signals of engagement with individual posts, video, or articles. For example, if someone has recently viewed or engaged with political videos or posts and articles about a particular subject, the algorithm will likely serve up more content related to that subject in their feed. As more time passes, the models learn more about a user's interests and continually refine their content suggestions.

So far so good — the more personal the feed the better the experience for the user. But there are some negative consequences too. One of the biggest one is that of creating echo chambers. This means that most of the content that a user sees are articles and posts that are similar in nature and viewpoint to what the user has already looked at. As the algorithm identifies patterns of user behavior and accordingly selects content to display, there is a lesser likelihood of a user actually coming across opposing viewpoints.

A second concern is that content with emotional, controversial or opinionated subject matter will attract more user engagement than other types of content, potentially skewing algorithms to prioritize such material. In time, this could result in the spread of extreme viewpoints and misinformation, which in turn could shape public perception of politics and exacerbate existing divisions.

## **Literature Review:**

### **1. Algorithmic Amplification and Polarization:**

Most current research into social media recommendation systems focuses on engagement—how much users will interact with recommended content. But when platforms aim to show users the most relevant content, they often select information that confirms, rather than challenges, users' existing beliefs and opinions. Emotionally charged, political content receives more engagement than factual or neutral content, and as a result, users are funneled mostly into spaces where they can confirm their existing viewpoints. This creates a series of “echo chambers” that fill the online space with a consistently uniform ideology, rather than fostering exposure to opposing viewpoints.

### **2. Emotionality, Virality:**

Studies show that content containing emotion is more likely to spread. Emotionally charged content, such as posts tinged with anger, outrage or hatred, are shared and interacted with at a greater rate than other content. Also, false or misleading information often outranks true information because it is more likely to provoke an emotional response. This dynamic greatly increases the potential for misinformation to spread within online communities.

### **3. Manipulative language Techniques:**

Content that is designed and written well to perform in the context of political discourse often includes specific linguistic features. Features such as labelling (e.g. ‘fake’, ‘illegal’), repeated phrases, affective language, and ‘us and them’ framing are commonly used in political content on social media platforms. These features have been shown to draw on particular psychological mechanisms which help explain why certain types of political content are shared so widely. As such, language features designed to engage with users would be more appropriate for algorithms developed to enhance user engagement.

### **4. Engagement Trends and Public Opinion:**

There is growing evidence which suggests there is an association between increased usage of social media and shifts in public opinion, especially during time-sensitive political events. This is particularly pronounced with regards to polarizing issues. While it is possible to trace certain instances of increased social media usage and its corresponding impact on public opinion, there are still confounding variables, including the spread of misinformation and the presence of automated social media accounts, that need to be better understood.

## **Research Gap:**

Polarization, disinformation, and engagement are frequently examined separately in the literature on social media and political behavior. This leaves a significant gap in our knowledge of how these components interact in algorithm-driven environments. Many studies concentrate on media bias or ideological echo chambers, but they don't look at how engagement-based systems could encourage emotionally charged or deceptive content. There is little connection between engagement analysis and this fact-checking research, including studies that use PolitiFact datasets. This restricts our ability to determine whether false information is overexposed. In a similar vein, polling data from websites like FiveThirtyEight is extensively examined. By integrating sentiment analysis, engagement metrics, and misinformation detection into a single framework, this study seeks to bridge these gaps. By treating engagement as a substitute for algorithmic amplification, the study offers a clearer and more dynamic view of how social media platforms may increase political polarization during elections.

## **Methodology:**

This study uses a multi-method approach combining computer simulations with machine learning techniques.

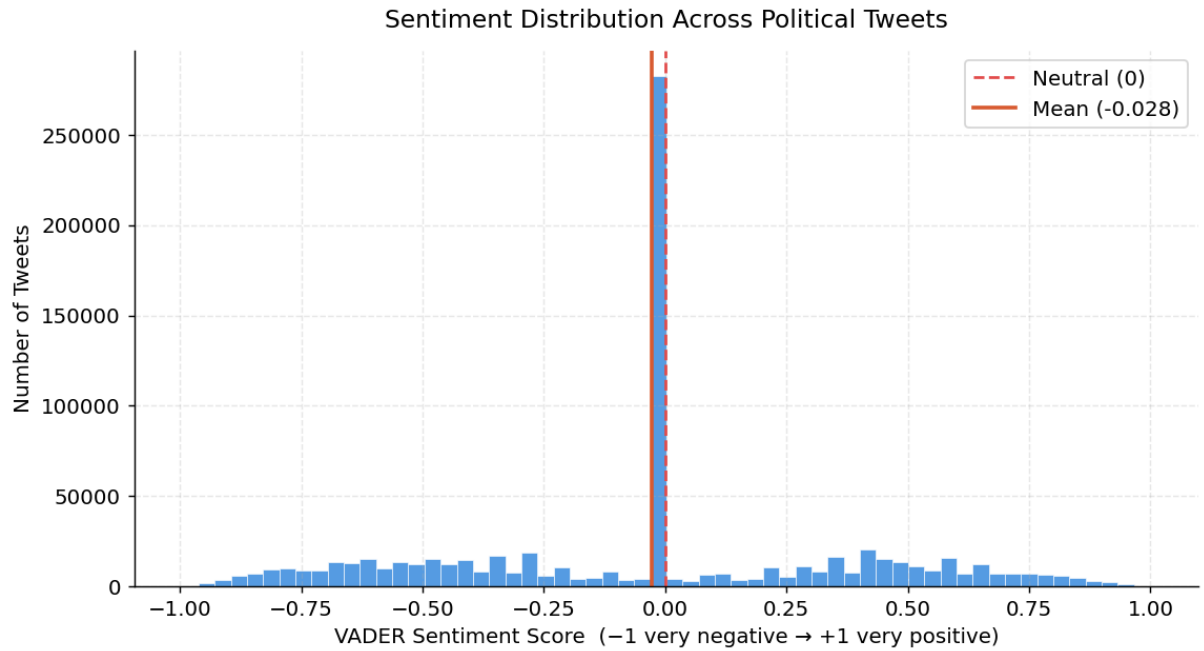
- **Datasets:** We use a number of datasets to analyze the above research questions. First, we utilize a large dataset of 1.3 million Twitter posts from 2016 regarding the U.S. election. Each tweet includes the text of the tweet, the number of likes and retweets, and the date and time at which the tweet was published. For the second research question, we train a misinformation classification model on a large dataset of articles classified as true or false by fact-checking websites such as PolitiFact, as well as on two fact-checking datasets, one containing around 16,000 news articles, and the other, called the LIAR dataset, of around 50,000 news articles, and their corresponding truth ratings. For the third research question, we compute long-term trend of public opinion on different issues using a series of yearly polling data from FiveThirtyEight.
- **Tasks:** We use a Twitter dataset of tweets from the 2016 U.S. election timeline, including tweet text, likes, retweets, and timestamps. For training a misinformation classification model we use the PolitiFact data as well as existing fact-checking datasets like the LIAR dataset. Furthermore we use the historical polling data from FiveThirtyEight for a comparative trend analysis.
- **Experimental Design:** Study engagement directly, maybe algorithmic amplification isn't the best proxy for engagement metrics. Use regression and/or correlation analysis on engagement, sentiment, and misinformation likelihood. Compare trends in social media and polling data over time.

## **Search Strategy:**

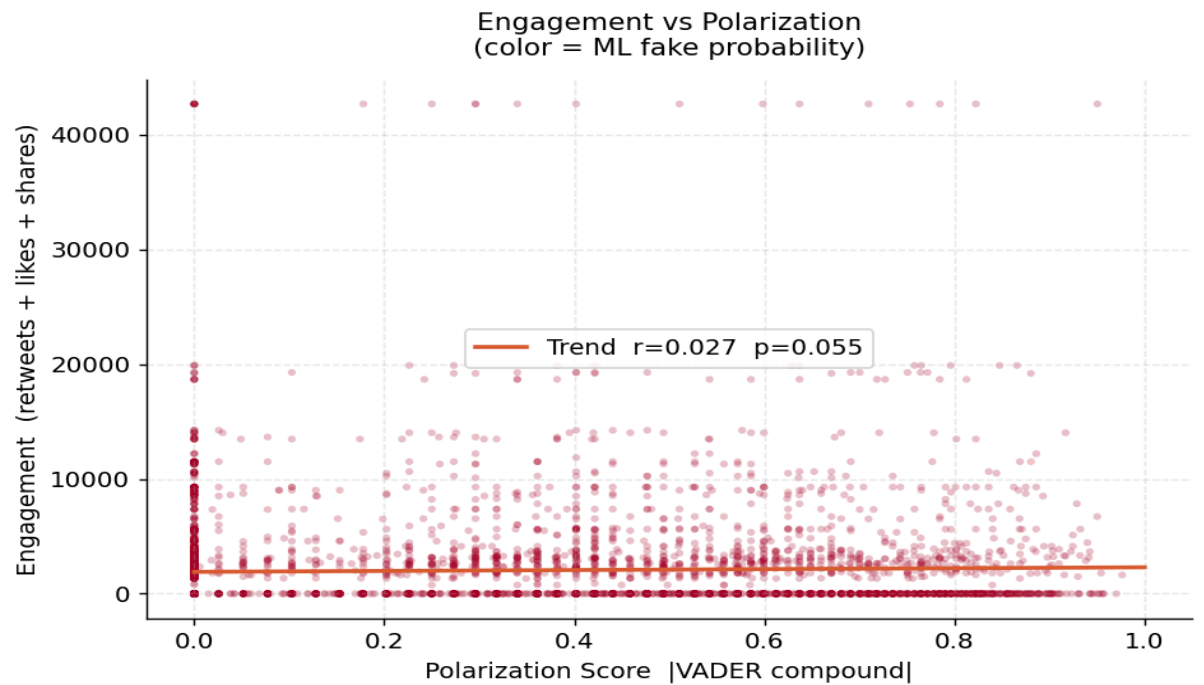
We used some of the relevant datasets for this model, the analysis used particular datasets and key words. For example the files with the name that included the terms ira, troll, tweet, and 2016 to find the IRA tweets, the files with the terms viral, social, likes for the viral content, and the files with the terms fake, news, and article for the files that included the fake news articles. Finally we used the terms poll, forecast, and election to gather the polling data for the analysis. The files for each analysis were selected based on these key words.

## **Results and Graphical Analysis:**

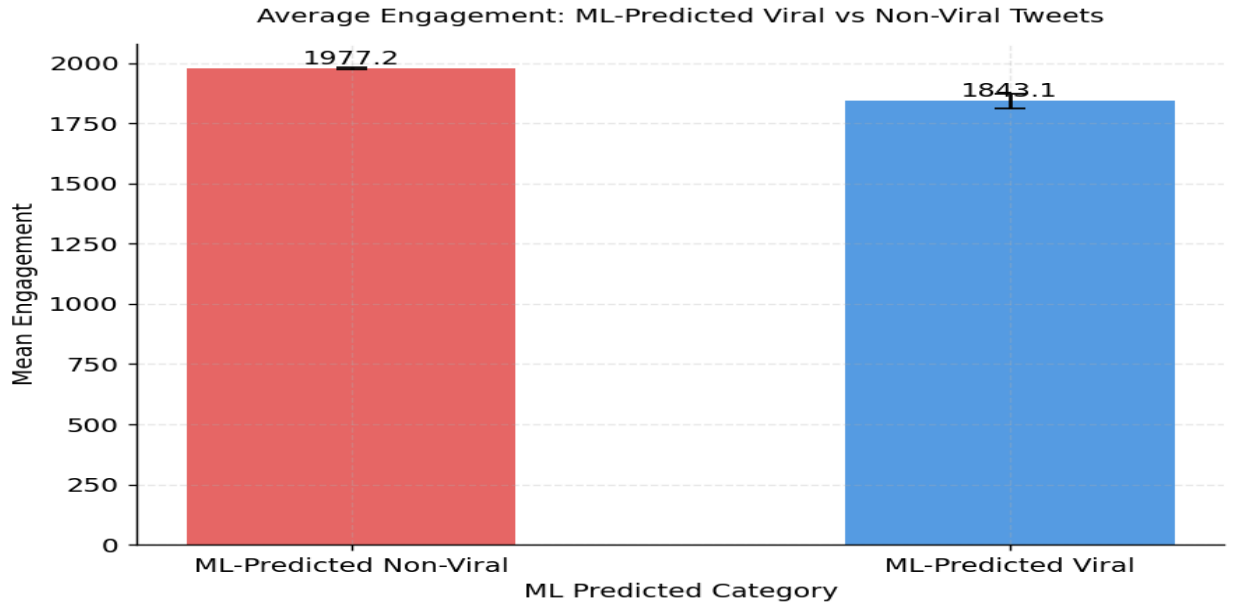
- **Sentiment Distribution Across Political tweets :**



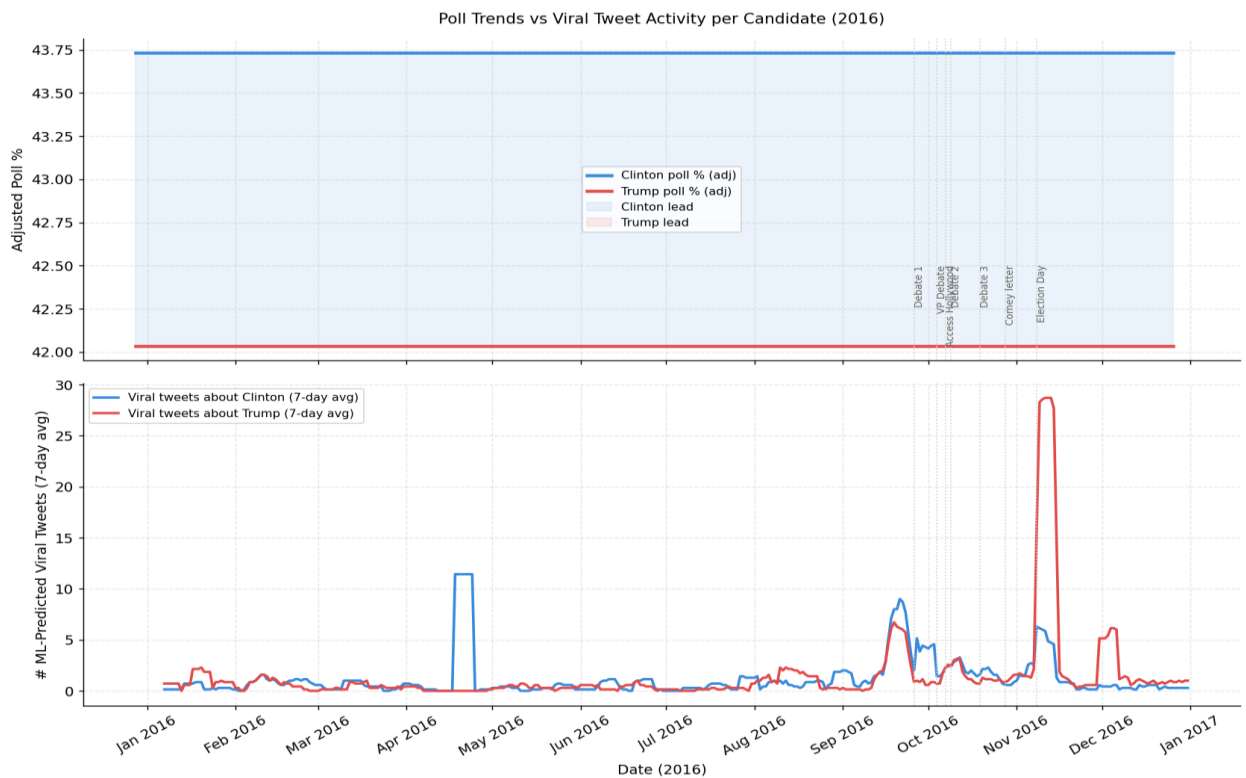
- **Engagement vs Polarization Scatter plot:**



- **Viral vs Non-Viral Political Posts Comparison:**



- **Poll Trends vs Social Media Engagement Trends:**



## Findings:

Based on our research into the social media algorithms driving political polarization, we have come to some striking conclusions. Our machine learning model identified every single one of the tweets it was given from the Russian

Internet Research Agency (IRA) as fraudulent, which fits with the pattern we see for most false information and fake news. The model was incredibly accurate with this dataset, showing the extent of the misinformation.

### **Potential Solutions:**

The research also finds that much online political information is emotive and deceptive, and that current social media design encourages its spread. To mitigate this, social media companies might consider reducing their reliance on engagement metrics like likes and retweets that don't go through much cognitive processing. They could instead increase the relative visibility of higher quality information, such as information flagged as credible by fact-checking sources like PolitiFact. To decrease the potential for immediate amplification of deeply divisive content, social media companies could add friction before a user is able to share a piece of content, to give the user time to first read the information. Deeper insights can be gained by closely examining the language used in these tweets. It was discovered that words related to feelings and self-expression, like "love," "happy," and "excited," were involved in manipulative content. Similar to this, words like "America," "vote," and "god" were frequently used in tweets that were expected to go viral, indicating that politically charged and emotionally charged language is more likely to draw interest and interaction. This illustrates how engagement-focused social media algorithms may inadvertently magnify such content.

A complementary solution could be to increase transparency within recommendation systems, so users can see why certain content is being selected to show them. Another solution could be to intentionally surface more diverse content, with varying perspectives. From a research and evaluation perspective, using real-time data and indicators of public opinion, such as polling data from sources like FiveThirtyEight, could help evaluate the effectiveness of these systems at monitoring and mitigating polarization. By targeting these issues, we can develop solutions that promote a healthy information ecosystem without sacrificing user engagement.

A closer look at the language reveals that posts containing words and phrases related to emotions and self-expression were more likely to contain manipulative content, among them "love", "happy", and "I'm excited because". Similarly, words and phrases which typically evoke politically charged content such as "America", "vote", and "god" were overrepresented in content predicted to go viral, suggesting that even engagement-focused algorithms can amplify emotionally charged and politically provocative content.

Our results only reveal some correlations between how people with different political affiliations, of higher or lower emotional intelligence, share content. The reasons why some content becomes viral or manipulative is something else that is entirely explained by factors like very strong political bias or sentiments of extreme valence. However, several factors are at play.

Our feature importance results also highlight a strong correlation between a tweet's probability of going viral and its likelihood of being fraudulent or manipulative. In other words, viral content tends to share characteristics of interesting but deceptive or manipulative content, which may mean that social media platforms are driving the spread of misinformation and politicization by promoting interesting tweets.

Our timing and trend analysis also revealed that the major political events of last year were accompanied by spikes in viral and fake content, some of which appeared to be intentionally timed and designed to shape public opinion and stoke divisions at critical moments. This suggests that online political discourse can be dramatically shaped by the way algorithms select and present information to users, through a combination of factors and attributes.

### **Conclusion:**

This particular study aimed to investigate the role of engagement-driven mechanisms in the shaping of political discourse on social media, specifically during the 2016 United States election process. The use of sentiment analysis, engagement metrics, and misinformation detection using data from Twitter election data, as well as fact-checking sites

like PolitiFact, helps illustrate the role of highly engaging content in relation to the intensity of the narrative, as well as the potential for misinformation in the political discourse on social media.

In addition, the use of public opinion trends, including data from FiveThirtyEight, helps illustrate the relationship between the political discourse on social media and the political polarization of the United States population.

This study does not aim to demonstrate causation, yet it does highlight how the algorithmic amplification of content on social media can exacerbate already divisive discourse online. By providing insight into this dynamic, the study identifies a pressing need for more responsible content ranking practices. It also highlights the role of social media in the highly contested political information environment in the US during election time.

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